Advancing 2D Monolayer CMOS Through Contact, Channel and Interface Engineering

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***Abstract*—2D CMOS transistors fabricated with transition metal dichalcogenide (TMD) materials are a potential replacement for silicon transistors at sub-12 nm channel length [L_G]. We demonstrate record NMOS contacts using a high melting point metal, down to 146 Ω-μm contact resistance (R_C). We present the best PMOS performance on a grown monolayer WSe_2 film with 50 μA/μm I_{on} and 141 mV/dec sub-threshold swing (SS) using a Ru contact metal, showing record PMOS contact resistance, R_C = 2.7 kΩ-μm. For the first time, we present 300 mm wafer growth options of 4 different 2D TMD films: MoS_2, WS_2, WSe_2, $MoSe_2$ that were grown at BEOL compatible temperatures. On unpassivated channel devices we show two methods of channel curing. First, N_2 desiccation can improve I_{ON} (~2×) and SS (~0.6×) simultaneously. Secondly, FGA annealing can improve bare channel devices by increasing their median I_{on} by 10× and lowering their SS by almost 50%. Finally, we benchmark our results against leading grown TMD devices, demonstrating record drive-currents among devices with good SS.**

I. Introduction

2D TMD transistors have a promising future as a CMOS replacement technology because TMDs do not suffer from channel scaling issues that limit Si transistors [1-2]. In sub 12-15 nm L_G Si CMOS, the Si channel needs to be aggressively scaled, which degrades the on-state mobility. 2D FETs do not suffer these intrinsic effects since their body thickness is already sub 1 nm without mobility loss relative to their bulk values.

However, 2D materials pose their own set of unique challenges – (1) the mobility of these materials is expected to be lower than Si and (2) Metallic contacts to 2D materials tend to be more resistive than metallic contacts to Si. To understand if these challenges impact the performance and power consumption of a transistor we performed a TCAD simulation of an NMOS stacked nanoribbon transistor. The results and the parameters chosen for the simulation are presented in Fig (1).

For the 2D transistor, (1) the number of ribbons is increased to six, (2) the L_G is scaled to 5nm and (3) the metallic contact replaces the epitaxial source/drain contact in a silicon transistor. Note that the increase in number of ribbons still results in the same overall gate height thereby avoiding extra parasitic capacitance. Our results show that even with assumptions of lower mobility (100 cm^2/V-s) and 2× higher contact resistance than Si, 2D transistor can achieve significant scaling, power and performance improvements. As the growth quality and integration of 2D materials improves one needs to take advantage of its unique scaling capability to offset any challenges that may arise.

In this paper we review the three key parts of the transistor in their own sections: **section II** channel material quality, **section III** contact resistance and **section IV** gate stack quality. In section II we outline two different growth techniques that are currently scalable, single crystal growth from a pre-patterned seed and MOCVD TMD growth on a 300 mm platform. In general, we find experimentally that MoS_2 NMOS FETs outperform WS_2 even though the theoretical mobilities are higher for WS_2. We also find that WSe_2 is the leading candidate for PMOS stacked nanosheets. In section III we will present a first of a kind PMOS contact that outperforms standard Pd contacts and the first experimental comparison between Au, Bi and Sb NMOS contacts which highlights promising new directions in understanding contacts to TMD materials. In section IV we show a technique for curing 2D transistor channels after processing and air exposure of unpassivated channels leading to improved performance.

The devices presented in this paper follow the integration process flow shown in Fig. (2) and are mostly fabricated from MBE then transferred Fig. (3) [3] [4]. The channel width and current normalization is done by the SEM measurement of channel width after etching. All process layers use standard e-beam processing. Metal was evaporated at $3e^{-6}$ Torr with a 15-minute prebake (125 °C in air) to drive off water. We believe this improvement is solely due to water removal since our unpassivated channel device performance improves by placing wafers in a 20 °C N_2 desiccator for 16 days, Fig (4). In this paper all measurements were performed in air. Using the above process, we fabricated an 18 nm 2D transistor, as seen in the TEM cross section Fig. (5)-(6).

II. Channel Material

The best data in literature, including our own data, for grown 2D FETs are devices fabricated from either solid source CVD or MBE, neither of which have been demonstrated at the 300 mm level to date. However, two methods of TMD synthesis do promise manufacturability at 300 mm scale: traditional heteroepitaxy on Si wafers by MOCVD and a novel nucleated CVD growth from pre-patterned seeds.

MOCVD has been used to synthesize compound semiconductors because it allows for precise control of precursor molar flow into a deposition chamber and is therefore a manufacturable process. A grand challenge for the MOCVD growth is the development of uniform 2D films with monolayer control and electrical quality comparable to silicon for relevant technologies [5] [6]. Here we share our first growths of MX_2 at the 300mm scale; M = Mo, W; X = S, Se), and note that our

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selection of organometallic precursors allows for a wide range of processing conditions, including temperatures as high as 1000 °C or down to 300 °C. This result opens the exciting option of BEOL 2D transistors. Fig (7a) includes identifiable Raman spectra for the four principal TMDs, and are taken from 300 mm wafers. The three images Fig. (7b, 7c, 7d) are taken from 300 mm wafers of different MoS_2 thicknesses, which explains the color differences and demonstrates thickness control. The Raman spectra offset vertically on each wafer are from center, middle and edge locations of the wafer showing excellent within wafer thickness control for all four TMD materials.

An alternative TMD growth approach which enables grain boundary free devices is a controlled nucleation approach or nucleated CVD growth from pre-patterned seeds. By pre-defining the metal oxide seed locations, we can restrict the TMD crystal location. This moves the metal source directly onto the wafer, skirting around the use of solid source metal oxides. By utilizing this nucleation approach and improving the process from our previous results [4] Fig. 8. We demonstrate high quality growth of monolayer WS_2 with an I_{ON} of 22 μA/μm, the best published result using nucleated seed CVD growth Fig. 9.

III. Contact Resistance

Low contact resistance to PMOS and NMOS remains a challenge for 2D FET, primarily due to Fermi level pinning of the contact and material interactions between the TMD and metal contacts [14]. Here we present two advances on electrically contacting TMD materials. First, we share our new NMOS contact material Sb with a relative comparison between Au, Bi and Sb contacts Fig. 10. Using the top 10 percentile of devices at each channel length (L_{CH}) we find Sb (R_C = 146 Ω–μm) outperforms Bi (R_C = 2.5 kΩ–μm) on MoS_2, although Bi with an equally low R_C on MoS_2 has been shown [7]. Because of its higher melting point, Sb has an intrinsic advantage over either Bi or Sn: (Sb = 631 °C, Bi = 271 °C, Sn = 231 °C). Therefore, it will survive BEOL temperatures, whereas Bi and Sn contacts will melt during BEOL processing.

PMOS contact advancement is far behind NMOS contact particularly with deposited 2D TMD films. Recent PMOS contact advances rely on exfoliated flakes [15]. In our experiments, we find Ru contacts are superior to both traditional Pd contacts and novel Cu contacts [8]. Figure 11 shows a grown monolayer WSe_2 FET with Ru contacts, it has a record drive current of 50 μA/μm I_{ON} for devices with good SS and I_{ON}/I_{OFF} ~ 7.5 orders of magnitude. The R_{total} vs L_G plot demonstrates a Ru R_C of 2.7 kΩ–μm. This is a significant step forward in making 2D PMOS devices on manufacturable 2D transistors. Additional challenge remains to successfully dope the contacts to TMD materials.

IV. Gate Stack Quality

2D van der Waals materials are well known to collect organic processing residues as well as adventitious carbon. Carbon-based surface contaminants can inhibit ALD nucleation and prevent high quality gate oxide growth, an already challenging process as 2D materials lack dangling bonds required for standard ALD growth. In Fig. 12, we compare 2D channel surface treatments, including a 400 °C 1hr vacuum anneal and a 300 °C 1hr forming gas anneal (FGA), for MoS_2 films directly grown and transferred onto an oxide substrate. XPS shows both anneals effectively reduce carbon contaminants. The additional carbon peaks detected for the transferred film from transfer process residues are also fully eliminated with the anneal treatments. The FGA treatment gives narrower FWHM Raman peaks corresponding to improved MoS_2 quality and enables gate oxide growth with oxide coverage reaching 100%.

In Fig. 13(a) a MOSCAP device with ML MoS_2 and AlO_x/HfO_2 bilayer shows large frequency dispersion in CV measurements before annealing. After the FGA the frequency dispersion is eliminated, indicating improved gate oxide quality and contact to MoS_2. In Figs. 13(b)-(d), IDVG curves for devices with transferred ML MoS_2 show 10× increase to I_{ON} and about 50% reduction in SS after FGA curing, demonstrating the importance of contaminant-free TMD channels.

V. Conclusion

By comparing data presented with the best published data, a clear picture of the state of 2D FET emerges. We present the data in both table (Table 1) and graphical (Fig 14) formats to give a clear view of the 2D landscape relative to Si target. For the benchmark we include only TMD FETs that meet the following criteria: FETs made from grown 1-3 monolayer films which can be transferred, $I_{on}/I_{off} > 10^6$ and SS < 1000 mV/dec. [9-13]. We included a very thick PMOS exception to the rule because of the lack of high current 2D PMOS on grown films [13]. The results show two key points. First the NMOS data is approaching silicon but still ~3× lower at target SS. Secondly, the PMOS data is half an order of magnitude behind NMOS, mostly due to contact resistance, and requires more research if TMDs are to replace Si CMOS.

In summary, we have shown new directions in both NMOS contacts [Sb contacts: R_C = 146 Ω–μm, melting point 631 °C] and PMOS contacts [Ru contacts: R_C = 2.7 kΩ–μm]. The new Ru contacts allowed us to achieve record PMOS FET performance on grown 2D TMD monolayer films with 50 μA/μm Ion and 141 mV/dec SS. We achieved BEOL compatible 300 mm MOCVD deposition of 4 different types of 2D TMD materials and a new record drive current for seeded nucleated growth of WS_2 with 22 μA/μm.

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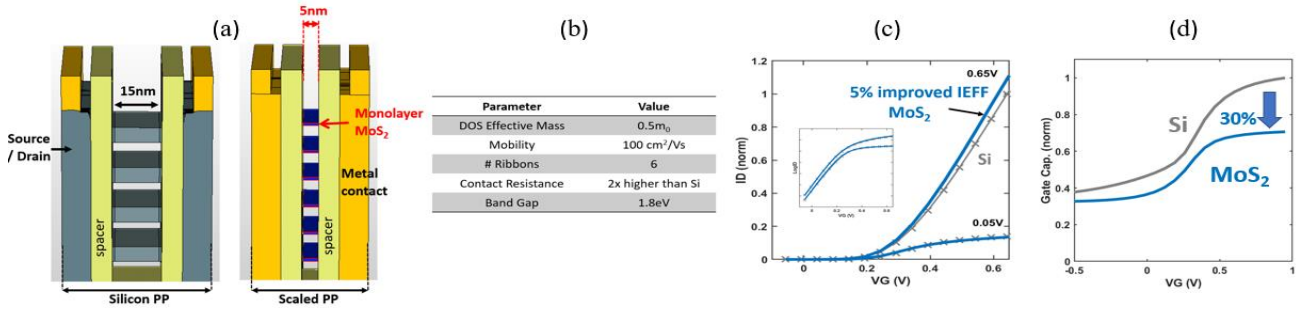


Parameter	Value
DOS Effective Mass	$0.5m_0$
Mobility	100 cm^2/Vs
# Ribbons	6
Contact Resistance	2x higher than Si
Band Gap	1.8eV

Fig 1. (a) Structure used to simulate IV and CV characteristics of Si and MoS_2 stacked nanoribbon NMOS transistor. The sub-nm channel in the MoS_2 case allows one to scale L_G down to 5nm and aggressively shrink poly-pitch. It also enables more ribbons to be stacked in the same vertical space. (b) MoS_2 parameters used in the Sentaurus device simulation. A 2D DOS with constant mobility was assumed. No changes were made in the gate stack relative to silicon case. (c) Assuming a mobility of 100 cm^2/Vs and 2× higher contact resistance for 2D TMD we still observed a 5% increase in ISEFF at Vcc=0.65V (d) Scaling the L_G down to 5 nm results in significant gate capacitance reduction of approximately 30%.

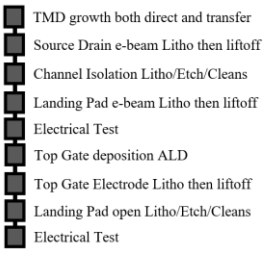


Fig. 2. Lab based wafer process flow.

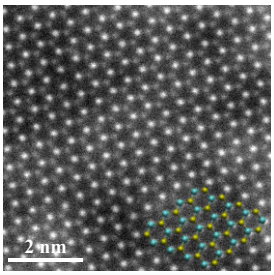


Fig 3. HAADF image of MoS_2 taken by ACSTEM. The brighter atoms correspond to a Molybdenum atom whereas the dimmer colored atoms correspond to two S atoms, as described in the ball-stick model inserted in the image. Acknowledgement JEOL Ltd.

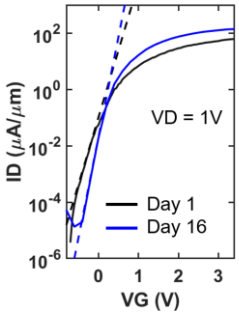
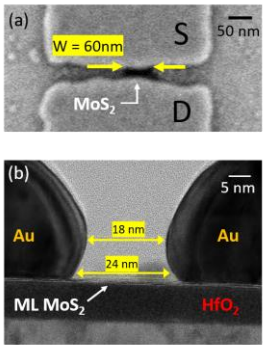


Fig. 4. Impact of a device sitting at room temperature in N_2 for 2 weeks, I_{ON} improves (64 mA/mm to 159 mA/mm), SS improves (210 to 140 mV/dec) and the Vt shifts +0.2 V.



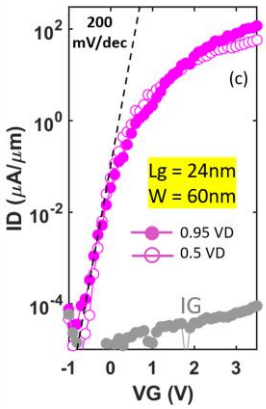


Fig. 5. (a) top down SEM of short channel device (b) cross section TEM of the same short channel device (c) I_DV_G of the same short channel device. High-κ dielectrics need to continue to scale to achieve target SS.

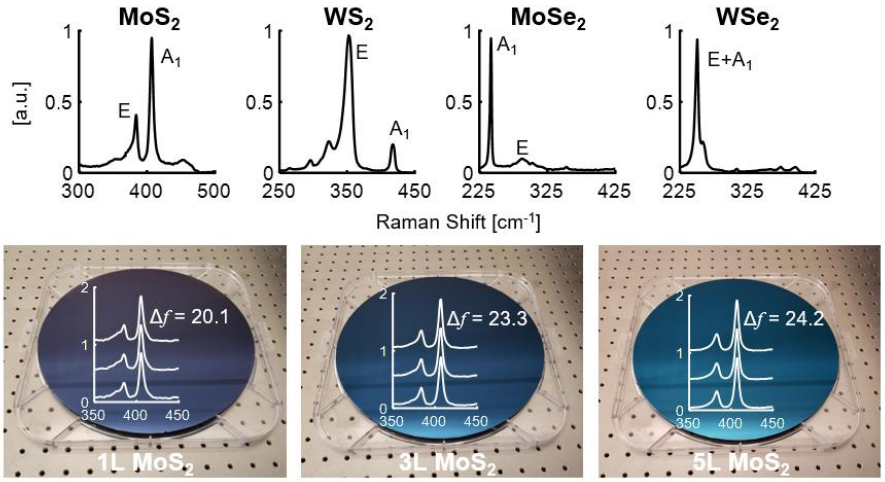


Fig. 7: (a) Raman spectra of the four TMDs grown on 300 mm wafers on 100 nm SiO_2/Si, detailing the characteristic E and A_1 modes among others for each material. All four materials are deposited at BEOL compatible temperatures in under one hour. (b, c, d) Optical images with inset Raman data demonstrating superb within-wafer uniformity and color change as a function of film thickness. MoS_2 is used as the example as the E-A_1 peak separation Δf directly correlates with film thickness. The three spectra shown in each inset are taken at the wafer center, mid-radius, and 1 cm from the edge.

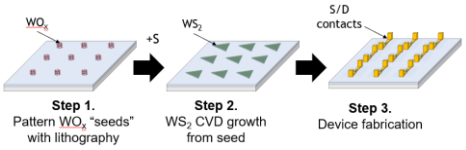


Fig. 8. Process flow for nucleated seed CVD growth.

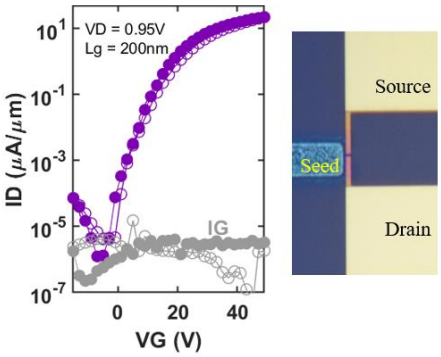


Fig. 9. Best WS_2 device made by the nucleated pre-patterned seed growth method. WS_2 grows from an on wafer WO_X seed achieving 22 μA/μm I_{ON} and I_{ON}/I_{OFF} ratio > 10^6. Device SS limited by the 100 nm SiO_2 back-gate. To the right we show an optical top down image of device.

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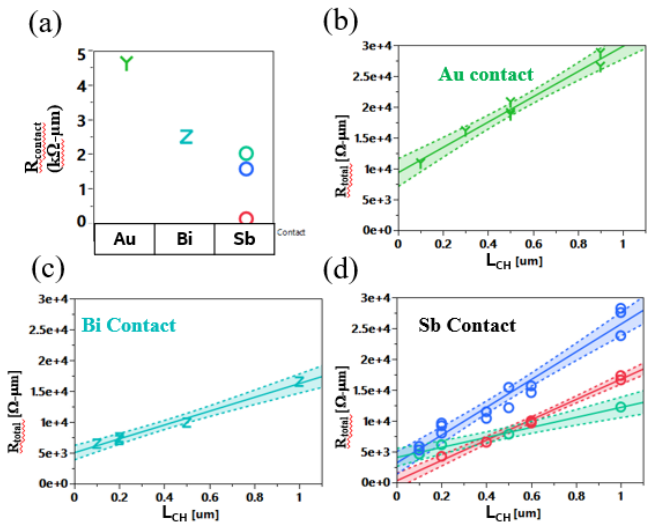

Fig. 10. NMOS contact comparison between Au, Bi and Sb. (a) Each point is the R_C of an individual wafer with MoS_2 film and different contact metal. R_{TOTAL} vs L_{CH} plots to solve for R_C (b) Au w/ R_C = 4.7 kΩ–µm, (c) Bi /w R_C = 2.5 kΩ–µm, (d) three wafers with **Sb contacts with best wafer R_C = 146 Ω–µm**, The plots are fits of the top 10 percentile data at each L_{CH} per wafer where V_D = 50 mV and $V_G = V_{OFF}$ + 2.5V

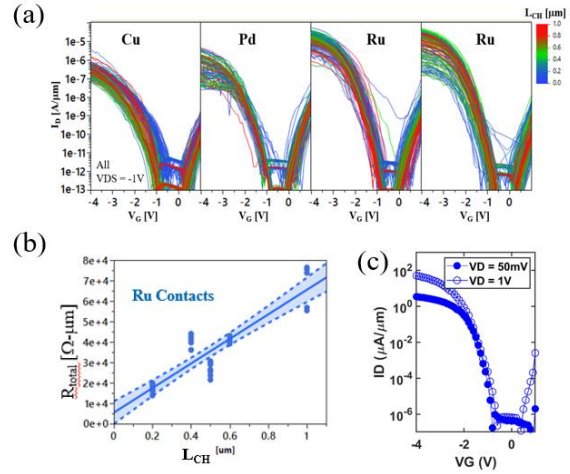

Fig. 11. (a) I_DV_G curves at -1V V_D for 4 different wafers with WSe_2 and Pd, Cu or Ru contacts, Ru has the highest I_{ON} in the group and the lowest R_C. (b) R_{TOTAL} vs L_{CH} plot of Ru contacts, R_C = 2.7 kΩ–µm, via fitting the top 25 percentile data at each L_{CH} where V_D = -50 mV and $V_G = V_{OFF}$ - 2.5V (c) I_DV_G for best device with **Ru contacts, 50 µA/µm I_{ON} and 141mV/dec SS**.

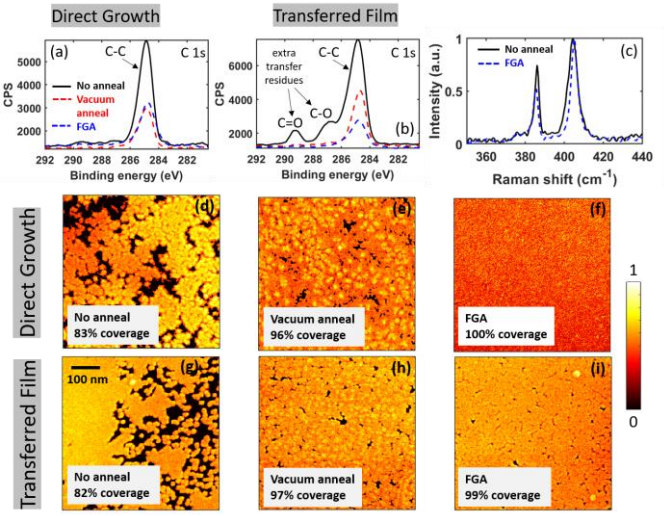

Fig. 12. (a)-(b) XPS shows removal of carbon residues through anneals. (c) Reduced FWHM from Raman spectroscopy shows improved MoS_2 quality post-FGA. (d)-(i) Al_2O_3 growth coverage on MoS_2 film increases to 100% with FGA.

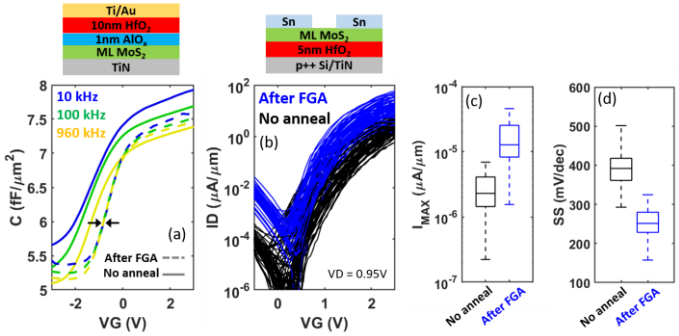

Fig. 13. (a) CV frequency dispersion reduced post-FGA for MOSCAP device with ML MoS_2 and AlO_x/HfO_2 bilayer. (b)-(d) I_DV_G curves for transferred ML MoS_2 show 10× increase to I_{ON} and ~50% reduction in SS after FGA cleaning.

Paper	Intel [4]	[7]	[9]	[10]	[11]	This Work	[12]	[13]
NMOS/PMOS	NMOS					PMOS		
Material	MoS2	MoS2	WS2	MoS2	MoS2	WSe2	WSe2	MoS2
# ML	1	1	~3	1	1	1	1	~7
Dep Method	MBE	SS CVD	MOCVD	SS CVD	SS CVD	SS CVD	SS CVD	CVD
Single or Dual Gate	single	single	dual	dual	dual	single	single	single - Ushaped
Imax (@ 1Vds) [µA/µm]	330	850	310	260	240 (@1.6Vds)	50	2	120
SS [mV/decade]	89	880	117	250	73	141	97	260

Table 1. Comparison table of device performance at 1V V_D for both NMOS and PMOS TMD FETs. Criteria for selection was grown or transferred TMD films of 1-3ML in all but one PMOS case. Further we only included FETs with I_{ON}/I_{OFF} ratios > 10^6 and SS less than 1000 mV/dec. In general, the PMOS devices lag the progress of NMOS devices.

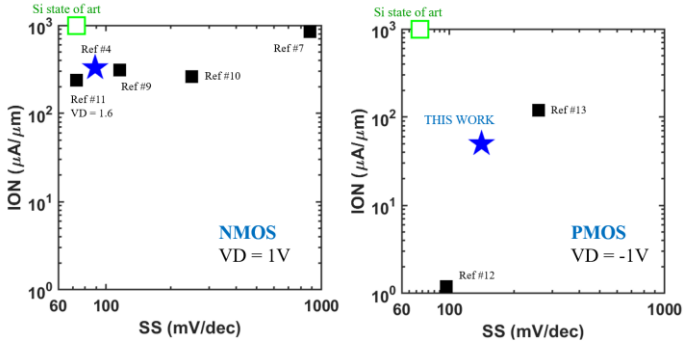

Fig. 14. Graphic of 2D benchmarking of devices from the Fig. 14. table using the same XY scales for ease of comparison. 2D TMD FETs must approach the Si state of the art to be competitive as a FEOL replacement.

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